PRODUCT REQUIREMENTS DOCUMENT

GYMTrack

An AI-Powered Workout Tracking and Progress Application

A student-built product requirements document designed for AI Product Manager portfolio evaluations, technical interviews, and product walk-throughs. This document explains the product rationale, user flows, AI grounding architecture, key product decisions, and future roadmap.

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Academic Background	Data Science & AI
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What a Recruiter or Interviewer will Learn from this Document:

- **Product Sense:** Why GYMTrack was built, who it serves, and what core gym friction it removes.
- **AI Product Thinking:** How the AI Coach is grounded in real historical workout telemetry to eliminate fake numbers and generic advice.
- **Prioritization & Decisions:** Concrete trade-offs made between MVP simplicity and advanced capabilities.
- **Execution & Challenges:** Real engineering obstacles encountered (local date sync, speech parsing, offline storage) and how they were solved.

1. Product Overview

GYMTrack is a fitness tracking application that allows users to record the exercises they perform, the weight they lift, and the number of sets and reps completed, and track their strength progression over time.

Alongside fast workout logging, GYMTrack includes an exercise library with video demonstrations, hands-free voice logging, progress analytics, offline-tolerant data caching, and an **AI Coach** that references the user's actual workout history to provide personalized, honest training guidance.

> **Core Philosophy:** GYMTrack prioritizes speed and honesty. It ensures that logging sets between exercises takes only seconds, and it guarantees that any feedback given by the AI Coach is strictly grounded in real recorded data rather than generic motivational text or fabricated numbers.

2. Problem Statement

Many people go to the gym regularly and work hard, but they fail to make consistent progress because they do not maintain an accurate record of their workouts. Specifically, lifters struggle with:

- **No Clear Workout History:** Lifters often cannot recall what weight or reps they achieved on their last workout session, leading to random guessing instead of systematic progressive overload.
- **Cumbersome Logging Interfaces:** Existing fitness apps often require navigating 5 to 7 screens, typing on tiny keyboards, or filling out long forms with sweaty or chalked hands during short rest periods.
- **Generic and Untrustworthy AI:** Most fitness apps with AI chatbots provide generic advice ('drink water and sleep 8 hours') or hallucinate personal records, making the guidance useless or even dangerous.
- **Lack of Visual Form Guidance:** Beginners and intermediate lifters often see confusing static drawings or broken links when trying to understand exercise mechanics inside the gym.
- **Poor Gym Network Reliability:** Basement gyms and metal-framed buildings frequently have weak cellular data, causing online-only apps to freeze or lose logged sets.

The Opportunity: Build a clean, focused web and mobile app that makes logging a workout effortless (in under 15 seconds) and turns that raw data into clear progression charts and grounded AI coaching.

3. Target Users

Rather than inventing fictional personas with synthetic demographics, GYMTrack is built around four real user profiles and their immediate training behaviors:

Beginner Gym Users

People who have recently started working out. They need clear visual exercise demonstrations to learn proper movement mechanics, safe starting weight suggestions, and an easy way to understand what muscle group to train next without feeling intimidated.

Regular Gym Users

Individuals who train 3 to 5 days per week. Their main requirement is speed: they want to record their sets quickly between exercises without interrupting their rest periods or getting distracted by social features.

Strength Progression Lifters

Lifters focused on progressive overload (adding weight or reps over time). They care about calculating their Estimated One-Rep Max (1RM), tracking volume trends across weeks, and seeing clear proof of strength gains.

Time-Constrained Lifters

Busy users who prefer speaking into their earphones ('bench press 60 kg for 8 reps, 3 sets') rather than typing numbers on a screen between heavy sets.

4. Product Goals

Goal	Success Description
Make Workout Logging Quick	Allow a user to log an exercise with weight, reps, and sets in fewer than 4 taps or with a single spoken voice sentence.
Maintain an Honest, Reliable History	Record every session accurately by local calendar date, ensuring no lost sets and zero synthetic data fabrication.
Provide Meaningful Progress Insights	Automatically compute daily volume, weekly trends, and estimated One-Rep Max (1RM) curves to make strength improvement visible.
Demystify Exercise Technique	Provide high-quality video demonstrations and execution cues directly on exercise cards so users do not need to switch to external search engines.
Deliver Grounded AI Guidance	Ensure the AI Coach answers user queries using their actual logged numbers, current goals, and recovery status rather than generic boilerplate responses.
Ensure High Reliability on Mobile	Ensure touch targets are optimized for one-handed phone use and local caching protects workout logs during gym network drops.

5. Main Features

GYMTrack organizes its core functionality into seven interconnected features designed to cover the complete workout lifecycle: preparation, active logging, post-workout reflection, and guidance.

Feature	What It Does	Why It Matters
Workout Logging	Allows users to record exercise name, sets, repetitions, and weight using rapid +/- steppers or direct number inputs.	Minimizes time spent tapping phone screens during active rest periods.
Exercise Library	A searchable catalog of exercises categorized by muscle group (Chest, Back, Legs, Shoulders, Arms, Core) with built-in video demonstrations.	Helps lifters verify form and learn new movements on the gym floor.
Progress Analytics	Interactive visual charts displaying total volume over time, workout session history, and estimated One-Rep Max (1RM) trends.	Provides visual proof of progressive overload and strength development.
AI Coach	A conversational training assistant that references the user's logged workouts, recent training frequency, and goals to provide personalized advice.	Gives lifters tailored answers (e.g., recovery, exercise alternatives) grounded in their real data.
Voice Logging	Speech-to-text logging that extracts exercise names, weights, sets, and reps from natural spoken language with an editable confirmation sheet.	Enables hands-free workout logging when users have chalk on their hands or wear gloves.
Rest Timer	A simple, customizable interval timer built directly into the workout session interface.	Helps users maintain consistent rest periods between sets to manage fatigue.
Structured Plans	Generates periodized multi-week programs (2M, 3M, 6M, 1Yr, 2Yr) with scheduled daily splits, inline video cues, and live checkmarks.	Eliminates workout guessing and ensures progressive overload across consecutive training blocks.
Offline Tolerance	Client-side caching (IndexedDB/localStorage) that saves workout records locally when cellular or Wi-Fi signals drop in gym basements.	Guarantees no lost workout data due to spotty connectivity; syncs automatically once reconnected.

Integration Note: All logged workouts automatically feed both the Progress analytics page and the AI Coach telemetry context in real time, ensuring that the system always operates on the most up-to-date user state.

6. User Flows

GYMTrack provides two complementary user flows for logging workouts: a standard manual logging flow for precise multi-set tracking, and a hands-free voice logging flow for rapid entry between sets.

Flow A: Standard Workout Logging Flow

This flow is designed for minimal taps and instant visual feedback on the home dashboard.

Step 1: Open App & Authenticate	User logs in with secure credentials; session is restored automatically on subsequent visits.
Step 2: Home Dashboard	User views today's completed sets, volume summary, and recent workout activity.
Step 3: Initiate Log Training	User taps 'Log Workout' button to open the quick-logging modal interface.
Step 4: Select Exercise	User selects target exercise from categorized dropdown or quick search (e.g., 'Barbell Bench Press').
Step 5: Adjust Weight, Reps & Sets	User adjusts values using rapid +/- incremental buttons or direct numeric input.
Step 6: Confirm & Save	User taps 'Save Workout'. Log is validated and committed to the database (and local cache).
Step 7: Real-Time Dashboard Update	Today's metrics increment immediately, progress charts update, and AI Coach context refreshes.

Flow B: Hands-Free Voice Logging Flow

Built for lifters with chalked hands or active earphone microphones during intense training.

1. Tap Voice Mic	User taps the microphone icon on the logging modal.
2. Speak Natural Sentence	User speaks: 'I did Incline Dumbbell Press 24 kg for 10 reps across 3 sets'.
3. AI Entity Extraction	System extracts: Exercise = Incline Dumbbell Press, Weight = 24 kg, Reps = 10, Sets = 3.
4. Editable Review Sheet	Extracted fields are displayed in an editable preview so the user can verify or correct numbers.
5. Confirm & Save	User taps confirm; workout is recorded identically to manual entry.

7. AI Coach: Grounded Intelligence

As an aspiring AI Product Manager, the design of the AI Coach represents one of the most critical parts of GYMTrack. In fitness technology, generic chatbots frequently fail because they lack personal context and hallucinate user accomplishments.

The Problem with Generic Fitness AI

Standard LLM integrations usually answer questions in a vacuum. If a user asks 'What should I train today?', a generic AI will output a standard generic split regardless of whether the user trained legs yesterday or has an injured shoulder. Similarly, when asked 'Am I progressing?', generic chatbots will guess or flatter the user.

How GYMTrack's AI Coach Works

GYMTrack solves this by using a **deterministic context injection pipeline**. Before any prompt reaches the AI, the application queries the user's database records and constructs a structured profile containing:

- **Recent Training History:** Exercises performed in the last 7 to 30 days, including weights, reps, and dates.
- **Muscle Group Frequency:** Hours elapsed since each major muscle group (Chest, Back, Legs, etc.) was last trained.
- **Strength Trends:** Estimated 1RM trajectory on key compound movements.
- **User Goals & Profile:** Training objective (Hypertrophy, Strength, General Fitness) and preferred units (kg/lbs).

Real-World Conversational Examples

User Query 1: ***'What should I train today?'***

AI Coach Response: 'Looking at your logged workouts, you trained Chest and Triceps 2 days ago (Bench Press 70kg for 3x8) and took yesterday as rest. Your Back and Biceps haven't been trained in 4 days. I recommend a Pull workout today: Barbell Rows, Lat Pulldowns, and Bicep Curls.'

User Query 2: ***'How do I perform a Barbell Squat correctly?'***

AI Coach Response: 'Here are the key cues: Keep your feet shoulder-width apart with toes slightly turned out. Brace your core, initiate by pushing your hips back, and keep your chest upright. Descend until hips are below knees. I have attached the verified Barbell Squat video demonstration from our library above for your review.'

User Query 3: ***'Am I getting stronger on Overhead Press?'***

AI Coach Response: 'Yes! Three weeks ago, you logged 40 kg for 6 reps (Estimated 1RM: 48 kg). Yesterday, you logged 45 kg for 5 reps (Estimated 1RM: 52.5 kg). Your strength on Overhead Press has increased by ~9.3% this month.'

8. Data & Personalization

GYMTrack's intelligence relies on clear, structured data models. By capturing only necessary workout parameters, the system maintains high data integrity while keeping user overhead to a minimum.

Data Inputs Collected

- **User Profile:** User ID, display name, primary fitness goal (Strength / Hypertrophy / General), preferred measurement unit (kg or lbs), and account creation timestamp.
- **Exercise Catalog:** Standardized exercise definitions: name, primary muscle group, secondary muscle groups, equipment type (barbell, dumbbell, cable, bodyweight), and video demonstration URL.
- **Workout Sessions:** Unique session ID, user ID, local calendar date, start timestamp, completion timestamp, and notes.
- **Set Logs:** Unique set ID, session ID, exercise ID, set order number, weight lifted, repetition count, and completion timestamp.

Derived Metrics & Calculations

Using the basic logged parameters (weight, sets, reps), GYMTrack deterministically calculates key performance metrics:

- **Daily Workout Volume:** Calculated as: *Sum(Weight × Reps)* across all completed sets for the selected day. Gives an instant measure of physical workload.
- **Estimated One-Rep Max (1RM):** Calculated using the industry-standard Epley formula: *1RM = Weight × (1 + Reps / 30)*. Enables lifters to compare strength across different rep ranges.
- **Muscle Group Frequency:** Tracks the number of sets dedicated to each muscle group over 7-day and 30-day windows to detect undertrained or overtrained muscle groups.
- **Workout Frequency & Consistency:** Tracks active workout days per week and consistency streaks without relying on punitive streak-loss mechanics.

The Data Honesty Principle:
A core requirement of GYMTrack is that the application **never displays fake workout history, simulated progress curves, or placeholder personal records**. If a user is new or has not logged workouts recently, the dashboard and AI Coach explicitly state that no data exists yet. This builds genuine trust and ensures that all visual progression represents authentic physical effort.

9. Key Product Decisions & Rationale

Product management is defined by the decisions made when balancing speed, user experience, and technical complexity. Below are seven deliberate product trade-offs made during the development of GYMTrack:

Product Decision	Context & Problem	Product Solution & Rationale
Decision 1: Rapid +/- Stepper Buttons for Logging	Lifters at the gym often have sweaty, chalked, or shaking hands between heavy sets. Typing numbers on a mobile keyboard causes errors and high friction.	Provided large, easy-to-tap +/- steppers that default to the previous set's weight and reps, reducing logging time to 2-3 quick taps.
Decision 2: Hands-Free Voice Logging with Editable Review	While voice logging eliminates keyboard typing entirely, speech-to-text models can occasionally mishear numbers or accented gym terms.	Implemented an editable review sheet where the AI extracts entities (exercise, weight, reps), but requires one explicit tap from the user before saving.
Decision 3: Daily Statistics Reset at Local Midnight	Users want to know exactly what they completed today. Using UTC timestamps caused late-night workouts (e.g., 10:30 PM) to split or count toward tomorrow.	Enforced local calendar date aggregation (00:00:00 to 23:59:59) so daily metrics reset accurately regardless of the user's timezone.
Decision 4: Separating Home (Today) from Progress (History)	Cramming long-term charts, historical logs, and today's session into one dashboard overwhelmed users and slowed down app load times.	Dedicated the Home screen strictly to today's active workout and immediate metrics, while dedicating the Progress screen to historical trends and charts.
Decision 5: Floating AI Coach Accessible from Any Screen	Users frequently need quick answers mid-workout (e.g., 'What is a substitute for Leg Press?' or 'How high should the bench incline be?').	Placed the AI Coach as an omnipresent bottom-right floating trigger so users can ask questions without navigating away from their active set log.
Decision 6: Embedded Video Demonstrations directly on Cards	When lifters do not know how to perform a movement, having to leave the app to search YouTube causes session drop-off and distraction.	Integrated high-definition video demonstrations directly into exercise detail cards with zero external app switching.
Decision 7: Strict Anti-Hallucination Guardrails on AI Coach	General chatbots tend to provide agreeable but fabricated answers when asked about past performance if data is sparse.	Engineered a system prompt that explicitly restricts the AI Coach: if an exercise or stat is not in the injected user context, the AI must admit it has no record.
Decision 8: Multi-Duration Periodization & Hybrid Persistence	Lifters plateau without structured progressive overload, while basement gym network drops break cloud-only multi-table plan creation.	Engineered multi-duration programs (2M-2Yr) with a dual Supabase + offline-first sync layer ensuring zero data loss and instant demo access.

10. MVP Scope vs. Later Enhancements

To avoid scope creep and validate core value quickly, the project was split into an initial Minimal Viable Product (MVP) and subsequent planned iterations.

Phase 1: MVP (Built & Validated First)	Phase 2: Later Enhancements (Added Sequentially)
• User Authentication (Email/Password) • Categorized Exercise Library (54+ exercises) • Manual Multi-Set Workout Logging • Rapid +/- Stepper Inputs • Home Daily Dashboard (Volume & Sets) • Basic Workout History & Session Logs • Grounded AI Coach with workout history context	• Hands-Free Voice Logging with speech extraction • Client-side offline caching & auto-sync • Interactive 1RM progression charts • In-app video demonstration player • Custom rest interval timer • Exercise search and muscle filter chips • Unit conversion toggles (kg/lbs)

11. Realistic Challenges & How I Solved Them

- **Dashboard State Synchronization:** Initially, logging a workout on the modal did not immediately update the Home screen's 'Sets Completed' counter until a full page reload occurred. *Solution:* Implemented a shared state listener / query invalidation system so that saving a workout triggers an instant reactive re-render of dashboard metrics.
- **Timezone & Date Skew:** Storing timestamps in standard UTC caused workouts logged at 10:00 PM local time to register under the next day's date on the daily summary. *Solution:* Standardized date handling by calculating the user's local day boundaries (midnight to midnight) before querying and aggregating daily workout logs.
- **AI Hallucinations on Missing Data:** When testing the AI Coach with a brand-new user account, the model invented past bench press records rather than acknowledging the account had no history. *Solution:* Enforced a strict system prompt constraint with zero-temperature instructions stating: 'If no workout history is provided in the context, explicitly inform the user that no workouts are recorded.'
- **Intermittent Gym Connectivity:** Testing in real gym environments revealed that basement weights rooms frequently lost mobile data, resulting in failed API network requests. *Solution:* Implemented a local IndexedDB buffer that saves the workout payload locally and syncs to Supabase once network connectivity is restored.
- **Mobile Thumb-Reach Accessibility:** The initial desktop-first UI had small buttons that were hard to tap with one hand on mobile screens while standing next to gym equipment. *Solution:* Redesigned the modal with 48px minimum touch targets, thumb-friendly bottom sheets, and high-contrast numerical indicators.
- **UUID Relational Schema Resolution in Plans:** Client exercises used string identifiers while database tables required UUID foreign keys, causing plan saving to fail with invalid input syntax. *Solution:* Engineered an automated UUID catalog resolution pipeline with backward-compatible local relational fallbacks.
- **Telemetry Scoping Without Redundant Columns:** Querying sets directly for user ownership failed because ownership resides on parent workout sessions. *Solution:* Refactored telemetry queries to join workout sessions with their child sets, ensuring accurate real-time completion tracking.

12. Success Metrics I Would Track

As an authentic student project, GYMTrack has not yet been deployed to a massive commercial user base. However, as a Product Manager, these are the exact qualitative and quantitative metrics I would instrument and monitor:

Metric Name	Measurement & Product Significance
Workout Logging Completion Rate	% of sessions where a user opens the logging interface and successfully saves at least 1 set. (Indicates logging ease and friction level).
Weekly Active Users (WAU)	Number of unique users who log at least one workout session per rolling 7-day period.
Average Logging Duration	Time in seconds between opening the log modal and confirming the set. (Target: under 15 seconds per set).
AI Coach Engagement Rate	Percentage of active lifters who ask at least one question to the AI Coach per week.
Voice Logging Adoption	Ratio of sets logged via voice vs. manual stepper inputs. (Indicates user appetite for hands-free interaction).
D7 and D30 User Retention	Percentage of users who log a workout in Week 1 and return to log workouts in Week 2 and Week 4.

13. Future Improvements

- **Wearable Device Integration:** Build an Apple Watch / Wear OS companion app for logging reps and checking rest timers directly from the wrist.
- **Smart Recovery Suggestions:** Automatically calculate muscle recovery fatigue scores to recommend optimal rest days before training the same muscle group.
- **Trainer Sharing & Export:** Allow users to export clean PDF progress summaries or share a view-only link with personal trainers.
- **Expanded Movement Demonstrations:** Add multi-angle exercise videos and dumbbell/machine alternatives for lifters training in crowded gyms.

14. Final Product Summary

Conclusion:
GYMTrack started as a simple workout tracking idea and evolved into a personalized, grounded fitness product. Its primary focus is making workout logging effortless while using the data collected from those workouts to provide meaningful strength progression analytics and honest, context-aware AI coaching. By eliminating unnecessary enterprise complexity and prioritizing user speed, data honesty, and mobile usability, GYMTrack demonstrates practical end-to-end product thinking from problem identification to functional implementation.